

Project Title

News Explorer

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

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ABSTRACT

Describe what your project presents within 150 words.

News Explorer is a searchable news repository designed to help users efficiently access relevant information from a large collection of news articles. With thousands of articles being published daily across different topics, finding specific and relevant information can become time-consuming. The proposed system addresses this problem by organizing news articles in a structured repository and providing efficient search and retrieval mechanisms.

Users can search for articles using **keywords, phrases, topics, or interests**. The system supports **keyword-based searching** for quick retrieval and **fuzzy matching** to handle misspelled or incorrectly entered queries. It also provides **topic-based filtering** to help users browse articles belonging to categories such as Sports, Technology, Politics, Business, and Entertainment. Furthermore, the system can identify and recommend **related news articles based on content similarity**.

The project focuses on applying Data Structures and Algorithms such as **KMP, Rabin-Karp, hashing, tries, and dynamic programming** to improve searching and retrieval efficiency. The final system aims to provide **fast, accurate, scalable, and user-friendly news exploration** while demonstrating the practical application of DSA concepts to a real-world information retrieval problem.

TECHNOLOGIES USED

Programming Language:	Java (JDK 17/21)
Environment:	IntelliJ IDEA / Eclipse / VS Code
Data Storage:	CSV/Text Files (or JSON files)
Version Control:	Git & GitHub
Algorithms	KMP, Rabin-Karp, Edit Distance (Dynamic Programming), Suffix Arrays (for similarity)

Signature of the Guide